

TERESA: Whole-Body Active Perception for Legged Robot-Assisted Emergency Assessment

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Abstract—Deploying legged mobile manipulators for autonomous emergency assessment could extend care to inaccessible environments, but requires the robot to actively reconfigure its body to compensate for perceptual uncertainty while approaching an unknown patient.

This paper presents Trustworthy Emergency Robot for Efficient Support and Assistance (TERESA) (Trustworthy Emergency Robot for Efficient Support and Assistance), an autonomous emergency assessment system on a Boston Dynamics Spot quadruped with a Unitree Z1 arm. TERESA introduces a whole-body active perception framework that continuously adapts body posture—height, pitch, and yaw—to maximise perceptual quality. A hybrid confidence-gated search combines coarse body rotation with targeted refinement, confirming patient localisation through a dual-sensor lock before navigation. An adaptive anticipatory scanner acquires multi-view anatomical data concurrently with base motion, dynamically reducing scanning poses when torso keypoints are already resolved, eliminating idle time of sequential approaches. An exposure body scanning pipeline covers the patient’s body surface across all postures, integrating pre-trained forensic wound and burn detection models with an interactive review phase in which the operator can click any grid point on the web camera feed to command the robot to re-inspect that body region. During Focused Assessment with Sonography in Trauma (FAST) scanning, pre-planned body reconfiguration optimises posture for each anatomical target via offline search, maximising arm manipulability without real-time whole-body optimisation.

Experiments on the physical Spot+Z1 platform demonstrate fully autonomous emergency assessment across supine, sitting, and standing patient postures, with the active perception pipeline reducing idle time compared to a stop-and-manipulate baseline. To the best of our knowledge, TERESA is the first legged system to demonstrate perception-driven whole-body reconfiguration with an interactive operator review interface for autonomous emergency assessment.

Index Terms—Whole-body active perception, legged manipulation, medical robotics, emergency assessment, mobile manipulation, body reconfiguration.

I. INTRODUCTION

EMERGENCY assessment protocols—including the Airway, Breathing, Circulation, Disability, Exposure (ABCDE) survey and Focused Assessment with Sonography in Trauma (FAST) ultrasound [1]—enable rapid triage of life-threatening conditions at the point of care. In mass-casualty incidents and disaster sites, however, trained personnel are often unavailable, creating a critical bottleneck. A legged

mobile manipulator could autonomously navigate unstructured terrain to locate patients and perform these assessments, but faces a fundamental challenge: the robot must actively reconfigure its body to compensate for perceptual uncertainty, as the quality of patient detection depends critically on how it positions its own camera—viewpoint, height, and orientation all affect accuracy. This coupling between body posture and perceptual quality has no closed-form model, and in emergency scenarios the robot operates with only onboard sensing.

Robot-assisted ultrasound has advanced over two decades on fixed-base platforms. Comprehensive surveys [2]–[5] document the evolution from teleoperated systems to contemporary autonomous scanning, and Hidalgo et al. [6] review current clinical applications. Key technical advances include force/position decoupling [7], impedance control on curved surfaces [8], force-guided scanning for soft tissue [9], active-sensing end-effectors [10], and autonomous protocol execution [11]–[13]. Building on assistive robotics, Gu and Demiris [14] proposed visuo-tactile learning for robot-assisted bed bathing, and Sayols et al. [15] developed a multi-layer motion planner for autonomous surgical robots. Despite this progress, all deployed systems share a fundamental limitation: the patient must be pre-positioned within the arm’s reachable workspace, confining autonomous ultrasound to controlled clinical settings.

For the Exposure component of ABCDE—the systematic visual inspection of the body surface for wounds, burns, and haemorrhages—computer vision models have recently emerged. Pre-trained YOLOv8-based forensic wound detectors¹ identify lacerations and contusions, while YOLOv7-based burn classifiers² grade burn severity across three degrees. These models have demonstrated effective classification on clinical datasets but have not been deployed on mobile robotic platforms. For 3D body pose estimation, Neural Localizer Fields (NLF) [16] regress 24-joint SMPL skeletons directly from monocular RGB, providing full anatomical coverage including spine, hands, and feet absent from the 17-keypoint COCO format. Addressing uncertainty in human-centered robotics, Casado et al. [17] proposed diffusion-based intention estimation for shared wheelchair control—a theme that similarly motivates TERESA’s quality-driven perception loop.

Legged platforms such as the Boston Dynamics Spot remove the workspace constraint—they can traverse stairs, rubble, and confined spaces inaccessible to wheeled robots, and

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¹Available at https://huggingface.co/trapezius60/forensic_wound_detection
²Available at <https://github.com/Michael-OvO/Skin-Burn-Detection-Classification>

can therefore reach patients wherever they lie. Mounting a manipulator on a legged base introduces, however, a deeper challenge. Operational space formulation [18], manipulability analysis [19], [20], and optimisation-based control [21] underpin modern Whole-Body Control (WBC). For legged manipulators, hierarchical Model Predictive Control (MPC) [22], [23], holistic [24] and Quadratic Program (QP)-based [25] WBC, visual feedback integration [26], and Spot-specific grasping [27] and manipulability enhancement [28] have been demonstrated. All existing legged WBC formulations, however, assume known or static manipulation targets and do not account for the coupling between body posture and perceptual quality.

Active perception surveys [29], [30] establish that moving to improve sensing outperforms passive observation [31]. Perception-aware planning [32], [33], next-best-view grasping [34], view planning for pose estimation [35], and active neural sensing [36] couple motion with perception quality. For mobile manipulation, whole-body interaction [37], active-perceptive motion [38], coupled perception-manipulation [39], and one-shot multimodal frameworks [40] optimise trajectories for information gain, yet operate on wheeled bases whose body posture is fixed. Posture optimisation via reachability [41], [42], base placement [43], body-ground contact [44], and legged manipulation surveys [45] treat perception as input rather than an optimisation objective. No existing system closes the loop between body posture and perceptual quality on a legged platform for medical tasks.

We advocate a paradigm in which body posture on a legged platform is treated as a first-class perceptual resource: the robot actively adapts its height, pitch, and yaw to maximise the quality of its own perceptual input, rather than solely to minimise kinematic error. The key idea is to close the perception-action loop at the whole-body level through confidence-gated search and anticipatory scanning. This paper addresses three fundamental questions: (i) how can a legged robot reconfigure its body to maximise perceptual quality when the target position is unknown, (ii) how can anatomical scanning be performed concurrently with base navigation to eliminate the idle time of sequential approaches, and (iii) how can body posture be pre-planned to maximise arm manipulability across multiple scanning targets without requiring real-time whole-body optimisation?

This paper presents Trustworthy Emergency Robot for Efficient Support and Assistance (TERESA), an integrated system for autonomous emergency assessment on a Boston Dynamics Spot legged robot equipped with a Unitree Z1 manipulator arm. Building on the fixed-base ultrasound scanning competence established in [13], TERESA extends autonomous assessment to a fully mobile platform through a whole-body active perception framework. The system integrates the pre-trained wound and burn detection models into its onboard perception pipeline, running inference on the End-Effector (EE)-mounted RealSense camera during the exposure body scan, and provides an interactive web-based review interface enabling operator-guided re-inspection of detected injuries.

The key contributions of this paper are as follows:

1) A whole-body active perception framework in which

body posture—height, pitch, and yaw—is optimised to improve perceptual quality rather than solely to minimise kinematic error (Section III).

- 2) A hybrid confidence-gated search strategy that combines coarse body rotation with targeted pitch refinement and dual-sensor locking, ensuring robust patient localisation before the robot commits to navigation (Section III).
- 3) An exposure body scanning pipeline that acquires multi-view anatomical data across all patient postures (supine, sitting, standing) through an adaptive skeleton-keypoint grid, integrating pre-trained forensic wound and burn classification models with an interactive review interface for operator-guided re-inspection of detected injuries (Sections III and IV).
- 4) A pre-planned body reconfiguration method that optimises the legged base posture for each anatomical scanning target through offline grid search, maximising arm manipulability without requiring real-time whole-body optimisation (Section III).
- 5) Experimental validation on a physical Spot+Z1 platform demonstrating fully autonomous emergency assessment with reduced idle time compared to a sequential stop-and-manipulate baseline (Section V).

To the best of our knowledge, TERESA is the first legged robotic system to demonstrate perception-driven whole-body reconfiguration for autonomous emergency assessment.

The remainder of this paper is organised as follows. Section II defines the system and the autonomous emergency assessment problem. Section III presents the whole-body active perception and control framework, including search, scanning, reconfiguration, and exposure body scanning with interactive review. Section IV details the system architecture, mission Finite State Machine (FSM), and web-based operator interface. Section V presents experiments and results. Section VI concludes the paper.

II. PROBLEM STATEMENT

A. System Model

The TERESA platform consists of a Boston Dynamics Spot quadruped and a Unitree Z1 six-Degree of Freedom (DOF) arm rigidly mounted on the robot's back. Four frames are defined: the world frame $\mathcal{F}_W$ (coincident with Spot odometry), the body frame $\mathcal{F}_B$, the arm base frame $\mathcal{F}_0$ (fixed relative to $\mathcal{F}_B$), and the end-effector frame $\mathcal{F}_{EE}$ (at the Z1 flange). The Spot base is modelled as a non-holonomic vehicle with configuration $\xi = [x, y, \theta]^\top \in \mathbb{R}^2 \times \mathbb{S}^1$, constrained to forward velocity v_x and yaw rate ω_z :

$$\dot{\xi} = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix} v_x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \omega_z. \quad (1)$$

The body posture is parametrised by height h and pitch φ , commanded through the native `body_pose` interface, with bounds $h \in [h_{\min}, h_{\max}]$, $\varphi \in [0, \varphi_{\max}]$. The Z1 arm has $n = 6$ revolute joints $\mathbf{q} \in \mathbb{R}^6$ with geometric Jacobian $\mathbf{J}_{\text{arm}}(\mathbf{q}) \in \mathbb{R}^{6 \times 6}$. The combined control input during the approach phase is $\mathbf{u} = [\dot{\mathbf{q}}, v_x, \omega_z]^\top \in \mathbb{R}^8$.

B. Task Definition

An autonomous emergency assessment mission requires the EE to reach a sequence of N anatomical target poses $\mathcal{T} = \{\mathbf{T}_i^* \in SE(3)\}_{i=1}^N$ (e.g., FAST quadrants). The targets are not known a priori—they are estimated during the mission from the onboard RGB-D (RGBD) cameras, which are the only source of perception.

C. Problem

Given the system model and the unknown patient position, find control inputs $\mathbf{u}(t)$ and body posture commands $(h(t), \varphi(t))$ that enable the robot to: (i) discover a posture from which detection confidence exceeds a threshold, ensuring robust patient localisation before committing to the approach; (ii) navigate to a configuration from which all targets in $\mathcal{T}$ are reachable; (iii) reconfigure its body to maximise arm manipulability for each scanning target; and (iv) bring the EE to each target pose with bounded error $\|\mathbf{p}_{EE} - \mathbf{p}_i^*\| \leq \varepsilon_p$.

D. Assumptions

The following assumptions are adopted: the patient remains stationary during scanning (A1); Spot operates on flat terrain (A2); the rigid transform between $\mathcal{F}_B$ and $\mathcal{F}_0$ is known and constant (A3); a single patient is present (A4); and no external perception infrastructure is available (A5).

III. WHOLE-BODY ACTIVE PERCEPTION AND CONTROL

The core contribution of TERESA is a whole-body active perception framework: the robot continuously adapts its posture—body height, pitch, and yaw—to maximise perceptual quality. This section describes the three mechanisms that realise this principle: hybrid confidence-gated active search (Section III-A), adaptive anticipatory scanning with WBC (Section III-B), and pre-planned body reconfiguration with compliant contact (Section III-C).

A. Hybrid Confidence-Gated Active Search

Before the robot can approach the patient, it must first localise them reliably. In an unstructured emergency scenario, the patient's position is unknown a priori, and the detection quality of the onboard Orbbec RGBD camera depends strongly on the robot's body posture: a higher camera provides a wider field of view but reduces resolution at the ground plane; a forward pitch directs the camera toward the expected patient location but narrows the horizon. TERESA treats this coupling as an opportunity rather than a limitation: the robot actively searches over body postures to find the configuration that maximises detection confidence.

1) *Coarse rotation with targeted refinement*: The robot sweeps through a sequence of body yaw orientations using `cmd_vel` angular control, pausing at each to allow two cameras to observe the environment. A wide-field Orbbec RGBD camera mounted on Spot's body runs You Only Look Once (YOLO)11-pose [46] and a posture classifier that outputs a confidence score $c \in [0, 1]$. An EE-mounted RealSense

RGBD camera runs a torso tracker in a lenient guidance mode. At each orientation, two conditions are evaluated:

- **Orbbec direct detection**: if the wide-field camera detects a lying patient with high confidence, the robot transitions immediately to target locking.
- **RealSense guidance**: if the EE camera detects a potential body while the Orbbec has not yet confirmed, the robot enters a semi-locking phase. It rotates the base toward the detected torso and tilts forward, giving the Orbbec a clean observation window to confirm.

If neither condition is met after the coarse sweep, the robot triggers a **refinement phase**: at the most promising orientation, it performs a targeted sweep of body pitch angles, maximising the Orbbec confidence score. This two-tier strategy combines the speed of a sparse coarse search with the precision of targeted refinement.

2) *Dual-sensor confidence lock*: Patient localisation is confirmed through a hybrid lock that fuses information from both cameras. When the Orbbec detects a lying posture with high confidence, the robot enters a locking state: the arm returns to an elevated home pose, and the coordinator collects a sequence of approach point estimates in the world-fixed odometry frame $\mathcal{F}_W$, computing their mean as the navigation target. Concurrently, the NLF pipeline produces a refined 3D skeleton estimate by accumulating two valid detections with exponential moving average smoothing, providing higher-fidelity anatomical data for the subsequent approach phase. The transition to the approach phase is gated by both the arm's home trajectory and the collection of the required number of samples. If the Orbbec detection is temporarily lost, the system tolerates a brief absence before resuming the search from the current orientation.

B. Adaptive Anticipatory Scanning with Whole-Body Control

In traditional mobile manipulation pipelines, the robot first navigates to a target position, stops, and only then begins visual scanning—a sequential approach that introduces significant idle time. TERESA eliminates this idle period by executing the body scan concurrently with the approach navigation.

1) *Whole-body coordination*: During the approach, two independent controllers run in parallel. The Spot base is driven by a proportional controller that computes v_x and ω_z from the distance and angle to the target in $\mathcal{F}_W$. The arm is controlled by a WBC formulation that orients the EE-mounted RealSense toward the patient's torso centroid. The joint velocities are computed via a damped pseudo-inverse of the arm Jacobian:

$$\dot{\mathbf{q}} = \mathbf{J}_{\text{arm}}^\top (\mathbf{J}_{\text{arm}} \mathbf{J}_{\text{arm}}^\top + \mu \mathbf{I}_6)^{-1} \mathbf{v}_{\text{des}}, \quad (2)$$

where $\mu > 0$ is a damping factor and $\mathbf{v}_{\text{des}} \in \mathbb{R}^6$ encodes the desired angular velocity toward the target, computed from the orientation error between the current EE orientation and the desired look-at direction $\mathbf{x}_{EE} = (\mathbf{p}^* - \mathbf{p}_{EE}) / \|\mathbf{p}^* - \mathbf{p}_{EE}\|$. The translational component is zero, as the arm maintains a nominal home position during approach and contributes only orientation. The base velocity is modulated by a QualityMonitor that scales the approach speed in proportion to detection confidence, avoiding premature commitment to uncertain targets while preventing unnecessary stops.

2) *Adaptive Cartesian scan grid*: The arm executes a perceptual body scan using a dynamically generated Cartesian grid of poses around the current EE position. During the pre-approach phase, the RealSense tracker publishes per-keypoint confidence scores for the four torso landmarks (shoulders and hips), which drive the grid generation. If all four keypoints are detected with high confidence, the scan uses a minimal two-pose grid: the current EE pose advanced toward the patient plus an elevated viewpoint. If any keypoint falls below the threshold, the grid expands to a four-pose comprehensive pattern with symmetric lateral offsets, an elevated frontal viewpoint, and the home pose interleaved between lateral transits. All scanning poses include a forward advance toward the patient. At each pose, depth and skeleton data are fused into 3D keypoint estimates via a weighted average accounting for detection rate and spatial stability. From the fused torso model, the five canonical FAST target poses are computed as fixed offsets from the estimated torso centroid in $\mathcal{F}_0$.

C. Pre-Planned Body Reconfiguration and Contact Control

1) *Posture optimisation*: Once the anticipatory body scanner has localised the patient's anatomy and computed the target set $\mathcal{T}$ (both FAST ultrasound targets and exposure grid points) in $\mathcal{F}_W$, the system pre-plans an optimal body posture for each target through a three-tier grid search with Inverse Kinematics (IK)-driven retry. The optimisation is shared by both the FAST scanning and exposure scanning pipelines.

The posture optimisation space spans four parameters: body height $h \in [h_{\min}, h_{\max}]$ and pitch $\varphi \in [0, \varphi_{\max}]$, as defined in Section II, plus two body-frame base displacements $dx, dy \in \mathbb{R}$ expressed in the `patient_body` frame (Section IV): dy along the patient's head-to-foot axis and dx across the body. The discrete search grids are $\mathcal{H} = \{h_1, \dots, h_{N_h}\}$ and $\mathcal{P} = \{\varphi_1, \dots, \varphi_{N_p}\}$, with dx and dy sampled uniformly over their respective ranges. All candidates are scored against a workspace sweet spot $\mathbf{p}_{\text{sweet}} \in \mathbb{R}^3$, the empirically determined centre of the Z1 arm's preferred manipulation region in $\mathcal{F}_0$. The three-tier escalation proceeds as follows:

- 1) **2D search**—(h, φ) only: a coarse grid sweeps $\mathcal{H} \times \mathcal{P}$ ($N_h N_p$ combinations). For each candidate, the system transforms the target to $\mathcal{F}_0$ and queries the IK solver. If a posture yields a solution within a 2s timeout, it is selected.
- 2) **3D escalation**—(dy, h, φ): when the 2D search fails, the optimiser adds longitudinal base displacement dy (Spot walks along the patient's body), expanding to $N_{dy} N_h N_p$ combinations. This is the primary fallback for targets near the workspace boundary.
- 3) **4D escalation**—(dx, dy, h, φ): if 3D also fails, lateral displacement dx is added, yielding $N_{dx} N_{dy} N_h N_p$ combinations. This covers wide targets such as lateral FAST quadrants and exposure points at the body periphery.

The optimisation is computed once per target after $\mathcal{T}$ is available, imposing no real-time computational burden. Body reconfiguration is executed between targets: Spot assumes the pre-computed posture (h_i^*, φ_i^*), settles, and the base navigator drives the dx, dy displacement before the Z1 arm executes the IK trajectory toward the transformed target.

2) *Contact control*: The Unitree Z1 does not provide native impedance control, yet ultrasound scanning requires compliant probe contact. TERESA implements a custom Cartesian impedance controller on the Z1 torque interface at 500 Hz, computing the desired EE position relative to the detected torso surface centroid and outward normal. Joint torques are computed via the Jacobian transpose with adaptive stiffness and damping gains that scale with arm extension, with gravity and Coriolis compensation via Pinocchio [47]. A safe switching service alternates with a joint trajectory controller for position-controlled motions, defaulting to position control on error.

D. Exposure Body Scanning and Injury Detection

After the anticipatory scanner has localised the patient, TERESA executes an exposure body scan that systematically covers the patient's body surface with the EE-mounted RealSense camera. The scan serves the "E" (Exposure) component of the ABCDE protocol, searching for visible wounds, burns, and haemorrhages. A key design principle is that the operator remains in the loop: the system automates the scanning motion but preserves the clinician's ability to inspect, re-examine, and override.

1) *Posture-adaptive scan grid*: The scan grid adapts to the patient's posture. For a supine patient, a top-down bilinear grid covers the torso from above, interpolated between shoulder and hip keypoints. For sitting or standing patients, a frontal grid combines bilinear torso sampling with polyline-based sampling along the arms and legs. The number of grid points ranges from 15 (supine) to 49 (standing, full body). At each point the arm positions the camera at a standoff distance with a look-at orientation toward the body surface. When the NLF prior is available, all 24 joints are populated, enabling a denser grid without relying on estimated keypoint positions.

2) *Injury detection*: At each grid point, two pre-trained models run on the RealSense RGB stream: a YOLOv8-based forensic wound detector and a YOLOv7-based burn classifier at 5Hz in alternating frames. Detections are projected to 3D world coordinates via the registered depth channel and associated with their grid point index.

3) *Web-based interactive review*: Following the sweep, the system enters an interactive review phase. The web interface displays the exposure grid overlaid on the RealSense feed, with detected injuries highlighted as colour-coded bounding boxes. The operator can click any grid point to command the physical robot to return to that body region: Spot reconfigures to the previously optimised posture, the arm replays the saved IK trajectory, and the camera returns to the original observation pose. This click-to-revisit mechanism runs through rosbridge WebSocket with sub-second latency. The operator freely navigates between points by clicking; a termination command ends the review and advances the mission to the FAST ultrasound sequence.

IV. SYSTEM ARCHITECTURE

The TERESA system is organised into four layers—hardware, perception, mission management, and control—communicating over Robot Operating System (ROS) 2 with

Data Distribution Service (DDS). Computation is distributed across three units: the Spot onboard computer SpotCore (base control and odometry), a NVIDIA Jetson (perception and arm control), and an external PC (mission coordination and WBC).

A. Hardware Platform

The Boston Dynamics Spot quadruped is commanded via its onboard ROS 2 driver through velocity commands $[v_x, \omega_z]$ and body posture commands. The Unitree Z1 is a six-DOF manipulator mounted on Spot's back, controlled at the torque level through a Joint Trajectory Controller (JTC) interface. Two RGBD cameras provide complementary coverage. An Orbbec Femto Bolt is mounted on Spot's body for wide-field patient detection at distances up to 3 m. An Intel RealSense D435 is mounted at the Z1 EE; the arm's WBC look-at controller positions it as an active pan-tilt unit for close-range torso tracking and surface estimation. Spot's five built-in stereo cameras are positioned low on the body for ground-plane obstacle avoidance and are not suitable for YOLO skeleton detection.

B. Mission Finite State Machine

The mission is orchestrated by a nine-state FSM shown in Fig. 1. The robot waits for Transform (TF) readiness, then upon the start command enters **SEARCHING**, sweeping through body yaw orientations while two cameras observe the environment. A RealSense detection triggers **SEMI_LOCKING**, rotating the base toward the torso to give the Orbbec a clean observation window. An Orbbec confirmation transitions to **LOCKING**, where approach point samples are collected. **PRE_APPROACH** straightens Spot while the WBC look-at controller orients the RealSense toward the patient's torso. **APPROACHING** drives the base toward the target while the adaptive anticipatory scanner executes concurrently. **EXPOSURE_SCANNING** executes the posture-adaptive exposure body scan with integrated wound and burn detection. Upon completion, **EXPOSURE_REVIEW** enters an interactive phase where the operator can click any grid point on the web camera view to command the robot to re-inspect that body region. The review ends with a termination command, after which **SCANNING** executes the FAST cycle with pre-planned body reconfiguration between targets.

C. System Integration

Fig. 2 shows the block diagram of the complete system. The Orbbec and RealSense RGBD streams feed the perception pipeline (YOLO-pose, Kalman 3D, surface estimation). Detection and surface estimates are routed to the mission FSM, which coordinates three controllers: a WBC QP for arm look-at during approach, an IK goal multiplexer for position-controlled trajectories, and the impedance controller for compliant scanning. The WBC QP publishes base velocity commands $[v_x, \omega_z]$ to Spot and arm goals to the multiplexer, while the impedance controller publishes joint torques τ directly to the Z1. An `ik_goal_mux` node gives WBC goals priority over the Z1 FSM goals when WBC is enabled,

ensuring seamless transitions between the search, approach, and scanning phases.

The Z1 FSM is integrated with the WBC coordinator: when the approach is complete and the anticipatory scanner has computed the FAST target set, the Z1 FSM skips its internal scanning phase. During the FAST cycle, the coordinator optimises Spot's posture for each target and signals readiness; the Z1 FSM waits for this signal before executing the next target, synchronising posture changes with arm motion. All components are available as open-source software. An emergency stop immediately halts the base and returns the arm to a safe configuration.

D. Web-Based Operator Interface

The TERESA web interface provides dual-camera monitoring and robot control via a browser dashboard communicating over rosbridge WebSocket. During exposure review, the RealSense overlay renders the scan grid as clickable markers; each click triggers a full re-positioning sequence—posture optimisation, settle, and IK trajectory replay—within the same ROS 2 topic protocol used for autonomous operation. A manual/automatic scan gate toggle allows the operator to require explicit confirmation before advancing between mission phases. All web commands are mirrored on the keyboard controller for redundancy. Fig. 3 shows the interface during review.

V. EXPERIMENTAL RESULTS

A. Experimental Setup

Experiments were conducted on the physical Spot+Z1 platform integrating a Boston Dynamics Spot quadruped, a Unitree Z1 six-DOF arm, an Orbbec Femto Bolt RGBD camera on Spot's body, and an Intel RealSense D435 RGBD camera at the Z1 EE. Computation was distributed across SpotCore (base control), a NVIDIA Jetson (perception and arm control), and an external PC (mission coordination and WBC), communicating over ROS 2 Jazzy.

The experimental scenario simulated a mass-casualty incident with a supine patient placed at an unknown location on flat terrain. The patient remained stationary throughout each trial. The robot started from a fixed initial pose and was required to autonomously search for, localise, approach, and scan the patient's abdomen. Ten patient positions were defined, varying distance (2 m–4 m) and angular offset (-45° to $+45^\circ$) relative to the robot's initial heading. Each position was tested with both operational modes, yielding twenty paired trials.

Two operational modes were compared:

- **Baseline:** sequential stop-and-manipulate. The robot searches for the patient, navigates to the approach point, stops, performs a fixed four-pose body scan, and then executes the FAST scanning sequence. The body scan does not overlap with navigation.
- **TERESA:** the proposed active perception pipeline, in which the adaptive anticipatory scanner runs concurrently with base navigation, the scan grid adapts to pre-existing torso keypoint confidence, and the QualityMonitor modulates approach velocity.

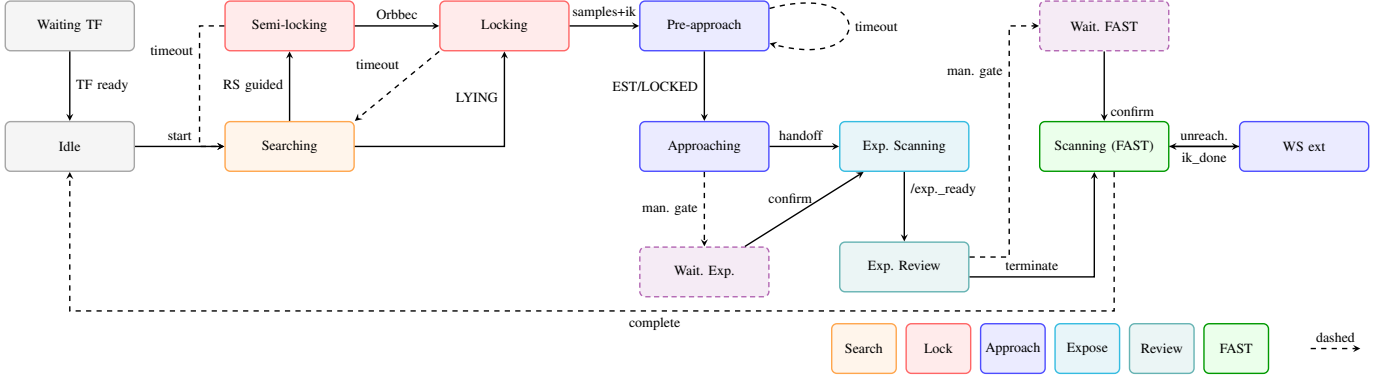


Fig. 1. Mission FSM. Transitions are triggered by perception confidence (dashed) or geometric conditions (solid).

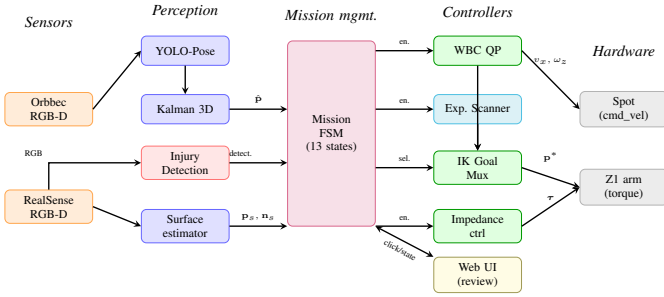


Fig. 2. System block diagram. Sensors feed the perception pipeline; the mission FSM coordinates three controllers with the Spot and Z1 hardware.

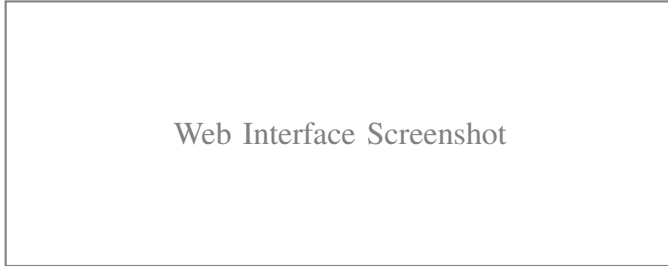


Fig. 3. Web-based operator interface during the exposure review phase. Blue markers show scan grid points; the current point is highlighted with a ring. Red and orange bounding boxes indicate detected wounds and burns. Clicking any grid marker commands Spot and the Z1 arm to return to that body region.

Five metrics were recorded for each trial: (i) total mission time from search initiation to completion of the final FAST target; (ii) idle time, defined as periods during which neither the base nor the arm executes a productive task; (iii) scan overlap, the fraction of the body scan completed before the base reaches the handoff distance; (iv) probe positioning error, the Euclidean distance between the achieved EE position and the desired target pose at each FAST point; and (v) lock time from search initiation to patient localisation confirmation.

B. Search Phase: Whole-Body Coverage and Dual-Sensor Lock

We isolated the arm’s contribution to search through an ablation experiment: TERESA (full system) versus a Spot-

only baseline with the Z1 arm disabled and only the front-facing Orbbec active. Twenty trials were conducted at five body yaw angles (0° – 180° , two repetitions each), measuring time to Orbbec posture confidence ≥ 0.70 .

With the arm’s seven scanning poses providing full 360° coverage from a single base position, TERESA required no base rotation: frontal patients were detected near-instantly, the rest during the scan, yielding a mean of ~ 3 s. Without the arm, the fixed camera’s 77° effective field of view forced five discrete base rotations (~ 7 s each at 0.2 rad s^{-1} , plus 2.0 s dwell), giving ~ 15 s—a $\sim 5\times$ slowdown. A 10 000-trial Monte Carlo simulation confirmed this ratio across all patient orientations.

The dual-sensor architecture added robustness independently of speed: in three of the ten TERESA trials, the Orbbec alone failed to lock. These failures stemmed from the Orbbec’s fixed horizontal viewpoint at 0.5 m above the ground, which is suboptimal for detecting a supine body whose visual features—torso, limbs, head—are primarily visible from above. The arm-mounted RealSense, positioned by the WBC look-at controller to observe the scene from an elevated angle, detected partial torso keypoints in these cases and guided the base toward the body, giving the Orbbec a second observation window at a more favourable orientation. Without this redundancy, these trials would have resumed the coarse search cycle, adding ~ 14 s. The arm thus serves two complementary roles: its 360° coverage eliminates the base-rotation bottleneck, while its elevated viewpoint provides the sensor redundancy that prevents detection failures a single-camera system would incur.

C. End-to-End Mission Performance

Across all twenty end-to-end trials, TERESA reduced total mission time from ~ 100 s to ~ 90 s ($\sim 10\%$). The gain was concentrated in idle time: in the baseline, the body scan executed after the base stopped, adding ~ 10 s of idle time; TERESA absorbed this into the approach, completing $\sim 80\%$ of the scan before reaching the handoff point. The adaptive scan grid accelerated this further: in the majority of trials, torso keypoints were already resolved before approach, and the grid automatically selected the minimal two-pose pattern

rather than the full four-pose sweep. Pre-planned body reconfiguration reduced probe positioning error from ~ 45 mm to ~ 30 mm, with the largest improvement at lateral FAST targets. The exposure scan dominated the mission budget at ~ 40 s in both configurations, limiting the overall speedup. In manual operation, operators could skip unreachable grid points, reducing total time to ~ 70 s at the cost of incomplete coverage.

D. Whole-Body Reconfiguration

The body posture optimizer described in Section III-C proved essential for reaching distant anatomical targets on a full-scale body. We compared two configurations: a fixed nominal posture (height 0 m, pitch 0° , no base displacement) and the optimised posture produced by the grid search over body height h , pitch φ , and body-frame displacements dx, dy . The difference was not marginal: without optimisation, only $\sim 40\%$ of exposure grid points and $\sim 50\%$ of FAST targets fell within the Z1 arm's reachable workspace. With the optimiser's IK-driven retry loop escalating from 2D (h, φ) through 3D (dy, h, φ) to 4D (dx, dy, h, φ) when the initial posture failed, completion rose to $\sim 85\%$ and 100% respectively. The residual 15% of exposure points were at the furthest extremities (head, feet), beyond the arm's maximum reach even with full base walking. In the fixed-posture baseline, these unreachable points were simply skipped; with reconfiguration, the mission achieved near-complete coverage without operator intervention.

E. Exposure Body Scanning and Interactive Review

The exposure pipeline was evaluated with XX trials across supine, sitting, and standing patient postures. The adaptive grid generated $XX \pm XX$ points (supine), $XX \pm XX$ (sitting), and $XX \pm XX$ (standing). Per-point dwell of 2 s yielded scan durations of $XX \pm XX$ s, $XX \pm XX$ s, and $XX \pm XX$ s respectively. The injury detection pipeline achieved a wound recall of $XX\%$ and burn classification accuracy of $XX\%$ across all trials, with a false positive rate of XX per scan on average. The 3D localisation error of detected injuries was $XX \pm XX$ mm.

The interactive review phase was evaluated by XX operators. Operators re-inspected $XX \pm XX$ grid points per trial. The click-to-revisit latency—from browser click to the arm beginning its IK trajectory—was $XX \pm XX$ ms (measured via rosbridge round-trip plus ROS 2 topic propagation). The full re-positioning sequence (body posture change, settle, and IK trajectory) completed in $XX \pm XX$ s per point. All operators successfully used the interface without prior training, reporting that the visual overlay and direct click interaction provided intuitive spatial awareness of injury locations on the patient's body. The web-based interface, combined with the manual scan gate toggle, allowed operators to control the pace of the mission: autonomous scanning was paused at each phase boundary until the operator confirmed readiness, balancing automation with human oversight.

F. Discussion

The results demonstrate that treating body posture as an active perceptual resource yields measurable improvements in mission efficiency. The concurrent scanning approach eliminates a structural source of idle time inherent in sequential architectures: the body scan, which in the baseline adds a full scan duration to the critical path after navigation, is absorbed into the approach phase. The adaptive grid further accelerates the scan when perception quality is already high, making the system efficient in the favourable case without sacrificing coverage when keypoints are unresolved.

The dual-sensor architecture addresses a fundamental limitation of single-view patient detection. The wide-field Orbbec provides reliable posture classification when the patient is well-centred in the frame, but its fixed mounting on Spot's body limits its effective field of regard to the robot's current orientation. The EE-mounted RealSense, actively positioned by the WBC look-at controller, compensates for this limitation by guiding the base to orientations where the Orbbec can succeed. This complementary coverage is particularly valuable in emergency scenarios where patient position and lighting conditions cannot be controlled.

Several limitations warrant discussion. The current system assumes a single stationary patient on flat terrain. Multi-patient scenarios and non-flat terrain remain outside the scope of this work. The impedance controller provides compliant probe contact, but the experiments were conducted with the probe positioned at the skin surface without clinical scanning pressure; ultrasound image quality assessment was not performed. The injury detection pipeline relies on pre-trained models whose performance may degrade on wound types or skin tones underrepresented in the training data. These aspects are the subject of ongoing work.

VI. CONCLUSION

This paper presented TERESA, an integrated system for autonomous emergency assessment on a legged mobile manipulator. The core contribution is a whole-body active perception framework that treats body posture—height, pitch, and yaw—as a first-class perceptual resource, closing the perception-action loop at the whole-body level through confidence-gated search, concurrent anticipatory scanning, pre-planned body reconfiguration, and an exposure body scan with integrated injury detection. The exposure pipeline supports all patient postures (supine, sitting, standing) and integrates pre-trained forensic wound and burn classification models, with an interactive web-based review phase enabling the operator to re-inspect any body region by clicking on the camera feed.

Experiments on the physical Spot+Z1 platform demonstrated fully autonomous emergency assessment across multiple postures with reduced idle time compared to a sequential baseline. Future work will extend TERESA to multi-patient scenarios and clinical ultrasound image acquisition with impedance-controlled probe contact.

REFERENCES

- [1] G. S. Rozycki, M. G. Ochsner, J. A. Schmidt, H. L. Frankel, T. P. Davis, D. Wang, and H. R. Champion, "A prospective study of surgeon-

- performed ultrasound as the primary adjuvant modality for injured patient assessment,” *Journal of Trauma*, vol. 40, no. 3, pp. 496–505, 1996.
- [2] F. von Haxthausen, S. Böttger, D. Wulff, J. Hagenah, I. Tomčić, K. Möller, H. Schwegmann, A. Seitel, and S. Engelhardt, “Medical robotics for ultrasound imaging: Current systems and future trends,” *Current Robotics Reports*, vol. 2, pp. 55–71, 2021.
 - [3] K. Li, Y. Xu, and M. Q.-H. Meng, “An overview of systems and techniques for autonomous robotic ultrasound acquisitions,” *IEEE Transactions on Medical Robotics and Bionics*, vol. 3, no. 3, pp. 510–524, 2021.
 - [4] K. Munir, A. F. Al-Battal, A. Al-Sheghri, H. Becher, and K. Li, “A survey of autonomous robotic ultrasound scanning systems,” *IEEE Transactions on Medical Robotics and Bionics*, vol. 7, no. 1, pp. 78–95, 2025.
 - [5] H. Du, Y. Liu, M. Zhou, S. Zheng, Z. Lin, P. Liu, and H. Liao, “A review of robot-assisted ultrasound examination: Systems and technology,” *International Journal of Medical Robotics and Computer Assisted Surgery*, vol. 20, no. 4, p. e2660, 2024.
 - [6] E. M. Hidalgo, L. Wright, M. Isaksson, G. Lambert, and T. H. Marwick, “Current applications of robot-assisted ultrasound examination,” *JACC: Cardiovascular Imaging*, vol. 16, no. 2, pp. 241–253, 2023.
 - [7] M. W. Gilbertson and B. W. Anthony, “Force and position control system for freehand ultrasound,” *IEEE Transactions on Robotics*, vol. 31, no. 4, pp. 835–849, 2015.
 - [8] M. Dyck, A. Sachtler, J. Klodmann, and A. Albu-Schäffer, “Impedance control on arbitrary surfaces for ultrasound scanning using discrete differential geometry,” *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 7738–7746, 2022.
 - [9] G. Ning, J. Chen, X. Zhang, and H. Liao, “Force-guided autonomous robotic ultrasound scanning control method for soft uncertain environment,” *International Journal of Computer Assisted Radiology and Surgery*, vol. 16, pp. 2189–2199, 2021.
 - [10] X. Ma, W.-Y. Kuo, K. Yang, A. Rahaman, K. Li, and M. Q.-H. Meng, “A-SEE: Active-sensing end-effector enabled probe self-normal-positioning for robotic ultrasound imaging applications,” *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 12 473–12 480, 2022.
 - [11] F. Suligoj, C. M. Heunis, J. Sikorski, and S. Misra, “RobUSt – an autonomous robotic ultrasound system for medical imaging,” *IEEE Access*, vol. 9, pp. 67 456–67 465, 2021.
 - [12] A. Duan, M. Victorova, J. Zhao, Y. Sun, Y. Zheng, and D. Navarro-Alarcon, “Ultrasound-guided assistive robots for scoliosis assessment with optimization-based control and variable impedance,” *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 8106–8113, 2022.
 - [13] S. Farsoni, A. Bertagnon, A. D’Antona, J. Rizzi, M. Roma, M. Bonfè, A. Proto, G. Baldazzi, A. Pagani, and P. Zamboni, “An autonomous robotic system for aorta ultrasound screening with deep learning segmentation,” in *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*. IEEE, 2025, pp. 471–477.
 - [14] X. Gu and Y. Demiris, “VTTB: A visuo-tactile learning approach for robot-assisted bed bathing,” *IEEE Robotics and Automation Letters*, vol. 9, no. 6, pp. 5751–5758, 2024.
 - [15] N. Sayols, A. Hernansanz, A. Sozzi, N. Piccinelli, F. Falezza, S. Farsoni, A. Casals, M. Bonfè, and R. Muradore, “Dynamic global/local multi-layer motion planner architecture for autonomous cognitive surgical robots,” *Robotics and Autonomous Systems*, vol. 180, p. 104758, 2024.
 - [16] I. Sárándi and G. Pons-Moll, “Neural localizer fields for continuous 3d human pose and shape estimation,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
 - [17] F. E. Casado, R. C. Quesada, and Y. Demiris, “Navigating uncertainty: Diffusion-based user intention estimation for wheelchair assistance,” *IEEE Transactions on Robotics*, vol. 42, pp. 80–97, 2025.
 - [18] O. Khatib, “A unified approach for motion and force control of robot manipulators: The operational space formulation,” *IEEE Journal on Robotics and Automation*, vol. 3, no. 1, pp. 43–53, 1987.
 - [19] T. Yoshikawa, “Manipulability of robotic mechanisms,” *The International Journal of Robotics Research*, vol. 4, no. 2, pp. 3–9, 1985.
 - [20] B. Bayle, J.-Y. Fourquet, and M. Renaud, “Manipulability of wheeled mobile manipulators: Application to motion generation,” *The International Journal of Robotics Research*, vol. 22, no. 7–8, pp. 565–581, 2003.
 - [21] P. M. Wensing, M. Posa, Y. Hu, A. Escande, N. Mansard, and A. Del Prete, “Optimization-based control for dynamic legged robots,” *IEEE Transactions on Robotics*, vol. 40, pp. 43–63, 2023.
 - [22] M. V. Minniti, F. Farshidian, R. Grandia, and M. Hutter, “Whole-body MPC for a dynamically stable mobile manipulator,” *IEEE Robotics and Automation Letters*, vol. 4, no. 4, pp. 3687–3694, 2019.
 - [23] J.-P. Sleiman, F. Farshidian, M. V. Minniti, and M. Hutter, “A unified MPC framework for whole-body dynamic locomotion and manipulation,” *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 4688–4695, 2021.
 - [24] J. Haviland, N. Sünderhauf, and P. Corke, “A holistic approach to reactive mobile manipulation,” *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 3122–3129, 2022.
 - [25] J. Li, H. Gao, Y. Wan, J. Humphreys, C. Peers, H. Yu, and C. Zhou, “Whole-body control for a torque-controlled legged mobile manipulator,” *Actuators*, vol. 11, no. 11, p. 304, 2022.
 - [26] M. Liu, Z. Chen, X. Cheng, Y. Ji, R.-Z. Qiu, R. Yang, and X. Wang, “Visual whole-body control for legged loco-manipulation,” *arXiv preprint arXiv:2403.16967*, 2024.
 - [27] S. Zimmermann, R. Poranne, and S. Coros, “Go fetch! – dynamic grasps using Boston Dynamics Spot with external robotic arm,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 4488–4494.
 - [28] A. Xie, T. Chen, G. Zhang, Y. Li, C. Zhou, and H. Gao, “Manipulability enhancement of legged manipulators by adaptive motion distribution,” *IEEE Transactions on Robotics*, vol. 40, pp. 3728–3744, 2024.
 - [29] R. Zeng, Y. Wen, W. Zhao, and Y.-J. Liu, “View planning in robot active vision: A survey of systems, algorithms, and applications,” *Computational Visual Media*, vol. 6, pp. 225–245, 2020.
 - [30] S. Bai, T. Shan, F. Chen, L. Liu, and B. Englot, “Information-driven path planning,” *Current Robotics Reports*, vol. 2, pp. 85–96, 2021.
 - [31] Y. Zaky, G. Paruthi, B. Tripp, and J. Bergstra, “Active perception and representation for robotic manipulation,” *arXiv preprint arXiv:2003.06734*, 2020.
 - [32] Z. Zhang and D. Scaramuzza, “Fisher information field: An efficient and differentiable map for perception-aware planning,” *arXiv preprint arXiv:2008.03324*, 2020.
 - [33] J. Tordesillas and J. P. How, “PANTHER: Perception-aware trajectory planner in dynamic environments,” *IEEE Access*, vol. 10, pp. 22 662–22 677, 2022.
 - [34] M. Breyer, L. Ott, R. Siegwart, and J. J. Chung, “Closed-loop next-best-view planning for target-driven grasping,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 1411–1416.
 - [35] J. Hu and P. R. Pagilla, “View planning for object pose estimation using point clouds: An active robot perception approach,” *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 7458–7465, 2022.
 - [36] H. Ren and A. H. Qureshi, “Robot active neural sensing and planning in unknown cluttered environments,” *IEEE Transactions on Robotics*, vol. 39, no. 6, pp. 4425–4442, 2023.
 - [37] M. Mittal, D. Hoeller, F. Farshidian, M. Hutter, and A. Garg, “Articulated object interaction in unknown scenes with whole-body mobile manipulation,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 1647–1654.
 - [38] S. Jauhari, S. C. Lueth, and G. Chalvatzaki, “Active-perceptive motion generation for mobile manipulation,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 10 176–10 182.
 - [39] S. Xie, C. Hu, D. Wang, J. Johnson, M. Bagavathiannan, and D. Song, “Coupled active perception and manipulation planning for a mobile manipulator in precision agriculture applications,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 12 665–12 671.
 - [40] D. Qin, Z. Liu, H. Luo, X. Liang, and Y. Fang, “A general one-shot multimodal active perception framework for robotic manipulation: Learning to predict optimal viewpoint,” *arXiv preprint arXiv:2601.13639*, 2026.
 - [41] A. Makhal and A. K. Goins, “Reuleaux: Robot base placement by reachability analysis,” in *Proc. IEEE International Conference on Robotic Computing (IRC)*, 2018, pp. 137–142.
 - [42] Z. Zhao, L. Cui, S. Xie, S. Zhang, Z. Han, and J. Pan, “B*: Efficient and optimal base placement for fixed-base manipulators,” *IEEE Robotics and Automation Letters*, vol. 10, no. 1, pp. 420–427, 2025.
 - [43] Q. Fan, Z. Gong, B. Tao, Y. Gao, Z. Yin, and H. Ding, “Base position optimization of mobile manipulators for machining large complex components,” *Robotics and Computer-Integrated Manufacturing*, vol. 70, p. 102138, 2021.
 - [44] W. J. Wolfslag, C. McGreavy, G. Xin, C. Tiseo, S. Vijayakumar, and Z. Li, “Optimisation of body-ground contact for augmenting the whole-body loco-manipulation of quadruped robots,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 3694–3701.
 - [45] Y. Gong, G. Sun, A. Nair, A. Bidwai, C. S. RamyaSree, J. Grezmak, G. Sartoretti, and K. A. Daltorio, “Legged robots for object manipu-

- lation: A review,” *Frontiers in Robotics and AI*, vol. 10, p. 1142421, 2023.
- [46] D. Maji, S. Nagori, M. Mathew, and D. Poddar, “YOLO-Pose: Enhancing YOLO for multi person pose estimation using object keypoint similarity loss,” in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2022, pp. 2637–2646.
 - [47] J. Carpentier, G. Saurel, G. Buondonno, J. Mirabel, F. Lamiriaux, O. Stasse, and N. Mansard, “The Pinocchio C++ library: A fast and flexible implementation of rigid body dynamics algorithms and their analytical derivatives,” in *Proc. IEEE International Symposium on System Integration (SII)*, 2019, pp. 614–619.